

Should anti-Müllerian hormone be a diagnosis criterion for polycystic ovary syndrome? An in-depth review of pros and cons

Emídio Vale-Fernandes,^{1,2,3,*}  Duarte Pignatelli,^{4,5,6,†}  and Mariana P. Monteiro^{2,3,†} 

¹Centre for Medically Assisted Procreation/Public Gamete Bank, Gynaecology Department, Centro Materno-Infantil do Norte Dr. Albino Aroso (CMIN), Centro Hospitalar Universitário de Santo António (CHUdSA), Unidade Local de Saúde de Santo António (ULSSA), Porto 4099-001, Portugal

²Clinical and Experimental Endocrinology, UMIB—Unit for Multidisciplinary Research in Biomedicine, ICBAS—School of Medicine and Biomedical Sciences, University of Porto, Porto 4050-346, Portugal

³ITR—Laboratory for Integrative and Translational Research in Population Health, Porto 4050-600, Portugal

⁴Faculty of Medicine, University of Porto, Porto 4200-319, Portugal

⁵Department of Endocrinology, University Hospital S João and Faculty of Medicine of the University of Porto, Porto 4200-319, Portugal

⁶I3S Institute for Innovation in Health, Porto 4200-135, Portugal

*Corresponding author: Unidade Local de Saúde de Santo António, EPE, Largo Professor Abel Salazar, Porto 4099-001, Portugal.

Email: emidiovalefernandes.pma@chporto.min-saude.pt

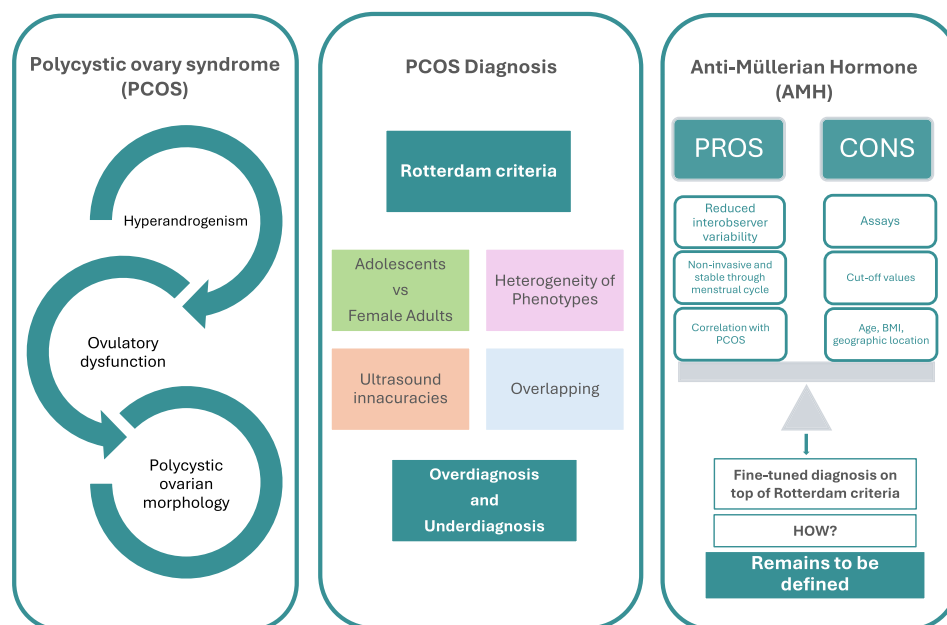
Abstract

Polycystic ovary syndrome (PCOS) is a prevalent endocrine disorder characterized by hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology (PCOM). Despite the widespread use of the Rotterdam criteria, challenges in diagnostic accuracy persist. Anti-Müllerian hormone (AMH), a glycoprotein secreted by ovarian follicles, has emerged as a promising biomarker for refining diagnosis due to its strong correlation with follicular count and elevated levels in women with PCOS. This review critically evaluates the advantages and limitations of incorporating AMH into PCOS diagnostic criteria. Elevated AMH levels are indicative of PCOM and anovulation, offering a non-invasive diagnostic tool that minimizes interobserver variability in ultrasound-based assessments. Additionally, AMH remains stable throughout the menstrual cycle and aligns with phenotypic diversity in PCOS, potentially supporting individualized management strategies. However, significant challenges remain. Variability in AMH assay methods, the absence of comparable cut-off values, and influences of age, ethnicity, and obesity on AMH levels limit its universal applicability. Additionally, AMH cut-offs for PCOS diagnosis, ranging from 3.5 to 5 ng/mL, raises questions about its clinical relevance, as there is not clear evidence of its biological significance. The review also highlights AMH's clinical utility in reproductive medicine, particularly in predicting ovarian response to stimulation, tailoring gonadotropin dosages, and optimizing assisted reproductive technology outcomes. While AMH holds promise as a complementary diagnostic criterion for PCOS, its fully integration into clinical practice requires further validation through standardized assays, population-specific cut-offs, and robust studies to address existing limitations. In conclusion, AMH harbours the potential to enhance the specificity and sensitivity of PCOS diagnosis, particularly in dubious cases. However, the inclusion of AMH in the current criteria for diagnosing PCOS still requires addressing methodological challenges and balancing its benefits against inherent limitations.

Keywords: AMH, ovarian function, PCOS, diagnosis of endocrine disease, infertility

[†] D.P. and M.P.M. contributed equally as last authors.

Graphical Abstract



Significance

Anti-Müllerian Hormone (AMH) is emerging as a promising biomarker for diagnosing Polycystic Ovary Syndrome (PCOS), a common endocrine disorder that significantly impacts women's reproductive and metabolic health. Unlike the Rotterdam diagnostic criteria that rely heavily on subjective assessments, AMH offers an objective measure of ovarian function tied to follicle count. Therefore, AMH inclusion into the current PCOS diagnosis criteria could enhance its accuracy. However, standardization of AMH quantification methods and establishment of population-specific cut-off values are still needed before clinical application can be considered. By refining PCOS diagnosis criteria, AMH could improve management strategies, and ultimately the global health of women affected.

Introduction

Polycystic ovary syndrome (PCOS) is one of the most common endocrine disorders in women at reproductive age, with an estimated prevalence ranging between 8% and 13% depending on the studied populations and accounting with up to 70% of undiagnosed cases.¹ PCOS is characterized by hyperandrogenism (HA), ovulatory dysfunction (OD), and polycystic ovarian morphology (PCOM), which can be associated with a variety of reproductive and metabolic disorders, including infertility, insulin resistance (IR), and an increased risk of developing type 2 diabetes and cardiovascular diseases.¹ Despite being highly prevalent, PCOS remains a complex condition with significant heterogeneity in clinical phenotypes, which represents a challenge to the establishment of universally accepted diagnostic criteria.²

The Rotterdam criteria were proposed by a panel of 27 experts, who issued a consensus document by the European Society of Human Reproduction (ESHRE) and the American Society for Reproductive Medicine (ASRM) in 2003, which nowadays is still the most widely used tool for diagnosing PCOS.³ According to the Rotterdam criteria, PCOS diagnosis is made when two of the following three features are present: (1) oligo- or anovulation, indicating irregular or absent menstrual cycles; (2) clinical or biochemical signs of HA, such as hirsutism, acne, or elevated androgen levels; and (3) polycystic ovaries, defined by the presence of 12 or more follicles in each

ovary or increased ovarian volume (>10 mL), as seen on ultrasound,³ providing other causes for these clinical features, such as congenital adrenal hyperplasia, androgen-secreting tumours, or Cushing's syndrome, have been excluded. The Rotterdam criteria acknowledge the heterogeneity of PCOS, even in women without overt HA or anovulation, by allowing for different phenotypes of the syndrome to be diagnosed, namely phenotype A (anovulation, HA, and polycystic ovaries), phenotype B (anovulation and HA), phenotype C (HA and polycystic ovaries), and phenotype D (anovulation and polycystic ovaries).²

Despite the fact that the Rotterdam criteria became the gold standard for diagnosing PCOS, this is not without limitations. First, the criteria do not consider the difference between adolescent and adult female physiology, and the overlap between PCOS symptoms and healthy puberty physiology results on PCOS overdiagnosis in adolescents.⁴⁻⁷ Another major criticism is the broad scope of each individual clinical feature considered by the criteria, which leads to significant heterogeneity in the phenotypes of women diagnosed with PCOS. By requiring the presence of only 2 among 3 possible features—oligo-anovulation, HA, and polycystic ovaries—the criteria allow for a wide spectrum of PCOS phenotypes.⁸ This can result in the overdiagnosis of milder forms of the syndrome, particularly in women with isolated polycystic ovaries but without HA or menstrual irregularity. Conversely, women

with HA and OD, but no visible polycystic ovaries on ultrasound may still be diagnosed with PCOS, creating ambiguity in phenotype definitions. Moreover, the presence of chronic anovulation and PCOM may also lead to PCOS diagnosis in women with hypothalamic amenorrhoea.⁹ In fact, the reliance on PCOM as one of the diagnostic criteria has proved to harbour several limitations.¹⁰ PCOM can be found in up to 25% of women with normal reproductive function, particularly in younger women, making PCOM less specific for PCOS.¹¹ In addition, the original cut off value of 12 follicles measuring between 2 and 9 mm of major diameter, which was established according to the available ultrasound technology at the time, is no longer valid. In fact, based on data obtained with modern imaging, it has been suggested to increase the threshold count of small ovarian follicles to at least 19 to 25.^{12,13} Additionally, in the context of PCOM, the variability in ultrasound technology and operator expertise can lead to inconsistent identification of polycystic ovaries, further complicating diagnosis.⁷ Moreover, the Rotterdam criteria do not consider the metabolic status nor the body mass index (BMI), namely the presence of IR or overweight and obesity, which are frequent in women with PCOS.^{8,14,15} Overall, the diagnostic process still relies on clinical signs, ultrasound findings, and biochemical markers, all of which have limitations in terms of accuracy and consistency across different populations. Therefore, depending on the individual characteristics, underdiagnosis, delayed diagnosis, and overdiagnosis of PCOS are possible. These challenges highlight the need for refinement of diagnosis criteria in order to improve the specificity and sensitivity of PCOS diagnosis, particularly when the symptoms are dubious.¹⁰ This would lead to a more accurate and timely diagnosis, allowing for targeted treatment and improved long-term outcomes for women with PCOS, in particular, the reproductive ones.

One biomarker that has garnered increasing attention over the years is Anti-Müllerian Hormone (AMH).^{16,17} AMH is a glycoprotein hormone produced by granulosa cells of small antral and pre-antral follicles, which serum levels are considered a reliable surrogate marker of ovarian reserve.¹⁸ Due to the increased follicular count, elevated AMH levels are commonly observed in women with PCOS.^{19,20} In this context, it has been suggested that AMH should be used as complementary criterion for diagnosing PCOS, especially when ultrasound findings are inconclusive or in younger women with less apparent or discrete symptoms.^{17,21,22} However, the incorporation of AMH into the diagnostic criteria for PCOS remains controversial and is a topic of ongoing debate considering multiple aspects and factors.^{7,23-25}

Given the evolving understanding of PCOS and the role of AMH, this review aims to explore the pros and cons of including AMH in the criteria for PCOS diagnosis. By examining the physiological role of AMH, its use as a diagnostic marker, its correlation with infertility, and its clinical applications, we herein provide a comprehensive overview of the current state of the art in this field. Our goal is to offer a balanced perspective on whether AMH should be included in PCOS diagnostic criteria, considering both the potential benefits and limitations.

Diagnosis and classification of polycystic ovary syndrome (PCOS) in women: a historical overview

What is now called PCOS is a medical condition that was first described in 1935, by the gynaecologists Irving Stein

and Michael Leventhal.²⁶ However, it took several decades before diagnostic criteria for the condition were proposed. The first diagnostic criteria for PCOS were issued in 1990, during a National Institutes of Health (NIH) conference, focusing on HA and OD, after excluding other endocrine disorders.²⁷ According to the NIH criteria, PCOS was defined as a hyperandrogenic disorder, which could present two different phenotypes: phenotype A (HA + OD + PCOM) and phenotype B (HA + OD). Using these criteria, the prevalence of PCOS in women at reproductive-age was estimated to be 5%-8%.²⁵

Despite providing the grounds for diagnosing PCOS, NIH criteria failed to encompass the full range of clinical features presented by the affected women, which led ESHRE and ASRM to issue the Rotterdam criteria in 2003.³ These criteria required the presence of at least two among three possible features for diagnosis: OD, HA, and PCOM, providing the exclusion of other endocrine disorders. The adoption of the Rotterdam criteria not only increased the estimated prevalence of PCOS to 8%-13% but also allowed to recognize additional phenotypes: phenotype C (HA + PCOM) and phenotype D (OD + PCOM).

The Rotterdam criteria were revised in 2006 by the Androgen Excess Society [now known as the Androgen Excess-PCOS Society (AE-PCOS)], by emphasizing the role of HA and recommending that PCOS should be defined by HA and OD, thereby excluding the non-hyperandrogenic phenotype D, which only presents OD and PCOM.^{28,29} However, these criteria were not widely accepted. In 2012, during the NIH Consensus Conference, the Rotterdam criteria were reaffirmed highlighting the need to document the various PCOS phenotypes and addressing the inadequacy of the term "Polycystic Ovary Syndrome", which could be misleading as the ovarian morphology is the end result of the expansion of antral follicles rather than the formation of cysts. Evidence-based guidelines have replaced consensus conferences for PCOS management, but the quality of evidence behind them is often low. Large-scale research initiatives to evaluate diagnostic criteria and treatment effectiveness are lacking, resulting in many recommendations based on consensus rather than strong evidence.³

The 2023 guidelines for diagnosing and managing PCOS stress the transition from consensus-based recommendations to those grounded in scientific evidence.⁷ These guidelines differ from the Rotterdam criteria, since these advocate for a nuanced, evidence-based approach to PCOS, including the use of AMH levels as an alternative to ultrasound in adults only, as well as patient-centred approach to PCOS. Namely, taking into consideration the long-term implications of PCOS diagnosis for its management, emphasizing the need to address metabolic factors, such as overweight/obesity and insulin resistance, alongside reproductive manifestations. Overall, these guidelines aim to enhance patient care by promoting a comprehensive understanding of the potentially associated health conditions and incorporating research recommendations to inform clinical outcomes.⁷ Therefore, the revised guidelines offer clinicians and patients well-defined recommendations on best practices, drawing from the most reliable evidence, insights from a diverse group of experts, and patient preferences. Additionally, the guideline is accompanied by research recommendations and a thorough dissemination and translation strategy, coupled with a comprehensive evaluation program to assess its effectiveness.

Role of AMH in ovarian function, reproduction, and infertility

AMH modulates the follicular response to follicle-stimulating hormone (FSH), thereby regulating folliculogenesis and ensuring the controlled progression of ovarian follicular development. As such, it plays a critical role in the regulation of ovarian function and reproductive physiology.³⁰

AMH remains relatively stable throughout the menstrual cycle and its serum levels progressively decline with age, paralleling the decrease in ovarian reserve.¹⁶ In women with PCOS, AMH levels are significantly elevated compared with women without PCOS, which is attributed to the greater number of small antral follicles present in polycystic ovaries³¹ and may contribute to the anovulation by inhibiting FSH sensitivity and follicular maturation.³²

The different levels of AMH observed in women with PCOS and without the syndrome has relevant clinical implications.^{33,34} Also, recent and ongoing research continues to elucidate the complex interaction between AMH and ovarian function,³⁰ particularly its role in reproductive outcomes³⁵⁻³⁷ and its potential as a therapeutic target in the management of PCOS.^{38,39}

Although high AMH levels in women with PCOS may suggest a greater ovarian reserve, the follicular environment is often dysfunctional, impairing the follicle ability to reach the ovulatory stage.⁴⁰ This results in chronic anovulation, which directly contributes to infertility. A recent study conducted by our group identified a positive correlation between circulating levels of AMH and follicular fluid (FF) oxidative stress (OS), which is known to impact on biological processes involved in the pathophysiology of several conditions associated with infertility, including PCOS, thus suggesting that AMH could act as a surrogate biomarker of FF OS in women with PCOS.⁴¹ This complex interplay between follicle quantity and quality, where elevated AMH levels reflect an underlying anovulatory state opens a possible route to be addressed in fertility treatments.^{42,43}

In this context, AMH became a key player to approach anovulatory infertility in PCOS, providing insights into ovarian function that go beyond traditional assessments, like ultrasound. In fact, it has proved to be relevant to assess the ovarian reserve, to predict the response to ovarian stimulation medications during in vitro fertilization (IVF), and to guide the selection of gonadotropin doses.^{44,45} Briefly, low AMH levels (< 1 ng/mL) indicate a low ovarian reserve, suggesting that the woman might respond poorly to stimulation, yielding fewer eggs for IVF.⁴⁵ In these situations, AMH levels are used as a guide for the adjustment of ovulation induction strategies.⁴⁶

On the other hand, high AMH levels (> 4-5 ng/mL) predict a strong response to ovarian stimulation.⁴⁵ However, this enhanced response can lead to an increased risk of ovarian hyperstimulation syndrome (OHSS), particularly in women undergoing controlled ovarian stimulation with gonadotrophins.⁴⁷ Monitoring AMH levels support clinicians to tailor gonadotropin dosages, minimizing the risk of OHSS while optimizing the chances of successful ovarian stimulation.⁴²

AMH levels are also useful in predicting the likelihood of success in achieving pregnancy, particularly in women undergoing fertility treatments.^{36,37,48} Higher AMH concentrations are associated with better oocyte yield during IVF cycles, which can enhance cumulative pregnancy rates.⁴⁹ However, despite an initial correlation between elevated AMH and

oocyte yield, it is crucial to recognize that these do not necessarily predict improved oocyte quality.⁵⁰ In fact, poor oocyte quality, despite abundant quantity, may still compromise overall fertility outcomes. This caveat limits the utility of AMH as a stand-alone predictor of treatment success and highlights the need for integrated approaches when managing infertility.⁵⁰

In a different perspective, the measurement of AMH may also be determinant for the choice of oocyte donor candidates.⁵¹ Indeed, our group has recently demonstrated that applying an AMH cut-off level of 1.12 ng/mL, predicted the yield of a number of eggs required for a donation to be considered successful, based on data consistently obtained across medical centres.⁵²

In parallel, AMH levels have been explored as a predictor of live birth rates in women with PCOS undergoing assisted reproductive technologies.^{53,54} While higher AMH levels generally suggest greater ovarian reserve and oocyte yields, the relationship between AMH and live birth rates is not as straightforward. Studies have reported contradictory findings regarding the predictive value of AMH for live birth, with some studies showing positive correlations between AMH levels and live birth outcomes,⁴⁴ while others demonstrate no significant association or even a negative association.^{53,55} The inconsistency stems from several factors. As referred, although AMH reflects the quantity of available follicles, it does not directly assess oocyte or embryo quality, which are critical for successful pregnancy and live birth. In addition, live birth rates are influenced by various factors beyond ovarian reserve, including age, sperm quality, uterine environment, and the presence of comorbidities such as IR or obesity, which are commonly associated with PCOS. Moreover, differences between study designs regarding population and sample size may also explain these disparities. More clinical studies and molecular biology experiments are warranted to elucidate on the relationship between AMH levels and pregnancy outcomes, as well as on the impact of AMH on folliculogenesis and oocyte quality in women with PCOS.

It should be noticed that AMH is not exclusively expressed in females, as it is also secreted by Sertoli cells in the testes and plays a paramount role in male reproduction. During embryonic development, AMH is involved in the sex differentiation by inducing the regression of the Müllerian ducts and therefore preventing the development of fallopian tubes, uterus, and the upper part of vagina, while promoting the development of male internal genitalia structures derived from the Wolffian tubes, including the epididymis, vas deferens, and seminal vesicles.⁵⁶ AMH levels decline gradually after birth until puberty but remain detectable throughout adulthood. However, its precise physiological role in adult males is still poorly understood. Notwithstanding, AMH levels in males can be used as a biomarker for detecting testicular tissue in conditions such as cryptorchidism or Sertoli cell-derived testicular tumours.⁵⁷

AMH as a diagnostic marker for PCOS

Multiple studies have proposed AMH as a potential diagnostic marker for PCOS and an opportunity to refine and fine-tune the Rotterdam criteria.^{16,58-60} One of the most widely referenced studies on this topic reported that women with PCOS had AMH levels up to three times higher compared with healthy controls.⁶¹ This correlation was attributed to the

larger number of small follicles in women with PCOS known to secrete greater amounts of AMH. The study concluded that AMH levels could effectively differentiate women with PCOS and those without the syndrome, particularly when symptoms were dubious or subtle. Sahmay and coworkers⁶² confirmed the link between AMH levels and PCOS, by demonstrating that AMH levels correlated positively with the severity of PCOS, especially in women with anovulation. Most authors suggested that AMH could not only serve as a diagnostic marker but also provide insights into the prognosis of PCOS, making it useful for tailoring treatment plans. However, a recent systematic review and meta-analysis, including 82 studies, concluded that AMH levels alone are insufficient for PCOS diagnosis and are nonspecific for PCOM in adolescents.¹⁷

Undoubtedly, AMH could potentially serve as a pivotal marker in the multifaceted approach to diagnosing and managing PCOS.⁶³ Indeed, the clinical relevance of using AMH levels as a reliable biomarker for PCOS, becomes particularly noticeable when compared with the subjectivity associated with transvaginal ultrasound.⁶⁴ Transvaginal ultrasound reliability, commonly used to visualize ovarian morphology and detect cysts or follicles, can be compromised by subjective interpretation and experience of the sonographer, leading to inconsistency in diagnosing PCOS.⁶⁵ This derives both from observer variability, who may report different findings based on their training and bias, and technical factors related to equipment quality and settings that can influence results.⁶⁶ The integration of AMH testing with transvaginal ultrasound could enhance diagnostic accuracy: AMH provides objective biochemical data that could complement subjective ultrasound findings and this combination could help to refine PCOS diagnosis.⁶⁷ How could this be effectively implemented, is a question that remains to be answered. Addressing this knowledge gap is essential for better understanding the role of AMH in PCOS diagnosis and improving clinical outcomes for affected individuals, overall highlighting the need for standardized AMH measurements.^{16,68}

AMH assays and implications for PCOS diagnosis

There are several methods available for measuring AMH levels, although the most frequently used are immunoassays, such as enzyme-linked immunosorbent assay (ELISA) or electrochemiluminescence immunoassay (ECLIA).⁶⁹ AMH immunoassays include manual ELISA assays (first and second generations), or automated ELISA assays (third generation), which consist in a one-step immunoenzymatic assay that uses two AMH-specific monoclonal antibodies in a sandwich format (Access® AMH, Beckman Coulter); while ECLIA immunoassays, can employ either a one-step sandwich immunoenzymatic method with fluorescence detection (Elecys® AMH, Roche; Vidas®, Biomérieux), or a two-step sandwich chemiluminescence approach (Lumipulse® G, Fujirebio).^{64,68,70,71}

Moreover, the coefficient of variation (CV) across different AMH assays is variable, which will also inevitably impact on the diagnostic accuracy of AMH for diagnosing PCOS. CV encompasses Intra-Assay CV that assesses the variability of multiple measurements taken within the same run, for which a CV < 10% is generally considered acceptable; and inter-assay CV that measures the variability between different runs and should ideally be under 15%. Understandably, achieving the lowest possible CV values is critical to ensure the most precise measurement of AMH⁶⁴ (Table 1). As such, high CV values can

undermine the reliability of using AMH as a diagnostic biomarker, as these indicate inconsistency in test results. Thus, standardization and method validation are crucial for clinical applicability of AMH as diagnosis criteria for PCOS.⁸¹ Additionally, different assays can also have different analytical sensitivity for detection, which determines the minimum levels that can be detected and consequently the lowest concentration of AMH can be accurately measured.⁶⁴ Indeed, AMH assays can have different lower detection limits, typically are in the range of 0.1 to 0.5 ng/mL, which may prove significant for diagnosis.^{64,70} Moreover, since AMH levels can be depicted in different units, either ng/mL or pmol/L, straightforward comparison of AMH levels across studies is not always possible and requires conversion (1 ng/mL = 7.14 pmol/L) prior to comparison and interpretation⁸² (Table 1).

AMH primarily exists in two isoforms: pro-AMH, the precursor form resulting from AMH gene transcription, and the short form, generated through proteolytic cleavage of pro-AMH. It is the short form that binds to the AMH receptor via its C-terminal region and is biologically active.⁷⁰ In the context of PCOS, this has a direct implication on the potential use of AMH as a diagnostic biomarker, since commercial AMH assays can either measure total AMH (pro-AMH + cleaved AMH), or specifically detect active AMH-C isoforms.¹⁶ Furthermore, whether the differential expression of AMH isoforms could be linked to specific clinical manifestations of PCOS or provide further insights into the diagnosis and management of PCOS also remains to be properly explored.^{16,70} Future research should focus on standardizing isoform specific AMH assays as well on elucidating the functional differences between these isoforms and their implications for diagnosis, as well as therapeutic strategies related to PCOS.

Given the abovementioned limitations of currently available methods for assessing AMH levels, it becomes clear that the development of standardized assays with low inter-method CV and detection limits is essential for enhancing the diagnostic utility of AMH. However, there is a notable lack of scientific data regarding the implications of different AMH assays for PCOS diagnosis. Currently available techniques, such as commercially available assays, vary in standardization and quality, which affects the reliability of AMH as a biomarker and despite its potential utility, the implications of different CV and minimum detection limits in the context of PCOS remains largely unexplored.⁸³

AMH and age bias

Research indicates that AMH levels typically decline with age, reflecting changes in ovarian reserve and function.⁸⁴ Several studies have documented patterns in AMH levels across different age brackets, with young women aged 15-25 years old exhibiting the highest AMH levels, generally stable, indicative of robust ovarian reserve and reflecting the peak of reproductive potential^{70,85}; middle-aged women aged 25-35 years old depicting a gradual decline in AMH levels as the number of primordial follicles decreases, more pronounced after the age of 30 years, while still retaining a relatively good ovarian reserve although fertility may start to decline^{70,86}; and pre-menopausal women aged 35-45 years old, typically showing a more accelerated decline in AMH levels, correlating with reduced fertility, with many women experiencing challenges with conception.⁸⁷

Overall, AMH can be highlighted as a crucial biomarker for assessing ovarian reserve and fertility potential, with age being

Table 1. Summary of proposed AMH cut-off values for PCOS diagnosis^a.

Reference	Type of study	Population	Cut-off for PCOS diagnosis (ng/mL)	Sensitivity for PCOS diagnosis (%)	Specificity for PCOS diagnosis (%)	Assay type	AMH assay		
							Limit of detection (ng/mL)	Coefficient of Variation (CV) % CV intra-assay	% CV inter-assay
Ahmed <i>et al.</i> , 2019 ²²	Case-control	Saudi Arabia	3.19	72.0	70.0	ELISA (Ansh Labs, Webster, TX, USA)	0.0012	2.07-3.69	3.15-5.84
Alhasan <i>et al.</i> , 2023 ⁷²	Cross-sectional	Sudan	3.95	99.0	100	ELFA (VIDAS® AMH; Austria)	0.01	<10.6	<10.6
Anand <i>et al.</i> , 2022 ²³	Meta-analysis	*	4.80	78.0	87.0	Among the 41 studies included, 22 studies used the Immunotech AMH assay (IOT assay, Immunotech-Beckman Coulter or DSL assay from Diagnostic systems laboratories), 11 studies used Gen II assay and 8 studies used automated assay (ROCHE, Dxi 600 assay of Beckman Coulter, VIDAS®)	*	*	*
Bell <i>et al.</i> , 2021 ⁷³	Cross-sectional	Australia	4.65	80.6	79.5	2-site immunometric assay (BA2 assay, Beckman Coulter, Queensland, Australia)	0.0154	1.5-1.7	3.5-3.6
Evliyaoglu <i>et al.</i> , 2020 ⁷⁴	Cross-sectional	Germany	6.16	80.6	84.8	ELISA (Ansh Labs, Webster, TX, USA)	0.0012	2.07-3.69	3.15-5.84
Wang <i>et al.</i> , 2024 ⁶³	Cross-sectional	China	5.33	89.0	96.0	Elecsys® AMH (Roche, Germany)	0.01	1.2-2.4	1.6-3.6
According to age (years old, yo)									
Barghi <i>et al.</i> , 2024 ⁷⁵	Meta-analysis	*	<19 yo: 6.00 19-25 yo: 5.96 25-30 yo: 5.52 ≥30 yo: 5.13 20-29 yo: 5.46 30-40 yo: 3.46 12-20 yo: 6.32 (including adolescents)	79.17 78.57 89.47 91.67 67.1 85.2 69.8	83.33 92.86 98.95 96.67 81.2 72.6 80.5	Automated Access® AMH and manual ELISA (49 studies) Automated/ELISA (149 studies) Access® AMH, Beckman Coulter Immunotech ELISA kit (Roche, Switzerland) Access® AMH (Beckman Coulter Japan, K.K., Tokyo, Japan) Lumipulse® G AMH (Fujirebio Diagnostics Japan, Inc., Tokyo, Japan) ECLusys® AMH Plus (Roche Diagnostics, K.K., Tokyo, Japan) known as Elecsys® AMH worldwide	*	*	*
Kakkad <i>et al.</i> , 2021 ⁷⁶	Cross-sectional	India	20-29 yo: 5.46 30-40 yo: 3.46	67.1 85.2	81.2 72.6	Access® AMH, Beckman Coulter Immunotech	0.02	<10.0	<10.0
Li <i>et al.</i> , 2022 ⁷⁷	Cross-sectional	China	12-20 yo: 6.32 (including adolescents)	69.8	80.5	ELISA kit (Roche, Switzerland)	0.01	1.2-2.4	1.6-3.6
Noguchi <i>et al.</i> , 2024 ⁷¹	Cross sectional	Japan	20-29 yo: 4.4 (Access/Lumipulse)/4.0 (Elecsys/ECLusys) 30-39 yo: 3.1 (Access/Lumipulse)/2.8 (Elecsys/ECLusys) <i>Rotterdam criteria</i> 20-29 yo: 9.9 (Access/Lumipulse)/9.0 (Elecsys/ECLusys) 30-34 yo: 7.3 (Access/	≥95	48-70	Access® AMH (Beckman Coulter Japan, K.K., Tokyo, Japan) Lumipulse® G AMH (Fujirebio Diagnostics Japan, Inc., Tokyo, Japan) ECLusys® AMH Plus (Roche Diagnostics, K.K., Tokyo, Japan) known as Elecsys® AMH worldwide	0.02 Access and Lumipulse are equivalent 0.01	<10.0 1.2-2.4 1.2-2.4	<10.0 1.6-3.6

(continued)

Table 1. Continued

Reference	Type of study	Population	Cut-off for PCOS diagnosis (ng/mL)	Sensitivity for PCOS diagnosis (%)	Specificity for PCOS diagnosis (%)	Assay type	AMH assay		
							Limit of detection (ng/mL)	Coefficient of Variation (CV)	
							% CV intra-assay	% CV inter-assay	
Tehrani <i>et al.</i> , 2021 ⁷⁸	Cross-sectional	Iran	Lumipulse/6.7 (Elecsys/ECLusys)						
			35-39 yo: 6.0 (Access/Lumipulse)/5.5						
			(Elecsys/ECLusys)						
			20-27 yo: 5.70	0.98 ^b	0.40 ^c	ELISA kit (Gen II, Beckman Coulter, Inc., Fullerton, California, USA)	0.08	1.9	2.0
			27-35 yo: 4.55	0.96 ^b	0.82 ^c				
Timur <i>et al.</i> , 2023 ⁷⁹	Cross-sectional	Turkey	35-40 yo: 3.72	0.86 ^b	0.96 ^c				
			21-25 yo: 5.56	80.65	75.00	AMH chemiluminescence immunoassay (Beckman Coulter, Brea, CA, USA)	0.08	3.1	3.1
			26-30 yo: 4.01	74.19	70.07				
			31-35 yo: 3.42	74.19	83.87				
According to BMI (Kg/m ²)									
Barghi <i>et al.</i> , 2024 ⁷⁵	Meta-analysis	*	<25 Kg/m ² : 5.93	79.55	96.59	<i>As quoted above</i>	*	*	
			≥25 Kg/m ² : 5.13	93.62	94.68				
Zhang <i>et al.</i> , 2023 ⁸⁰	Retrospective	China	<18.5 Kg/m ² : 5.145	84.3	89.1	Elecsys® AMH (Roche, Germany)	0.01	1.2-2.4	2.6-3.6
			18.5-24 Kg/m ² : 4.345	83.5	84.2				
			24.0-28 Kg/m ² : 4.115	81.0	84.2				
			≥28 Kg/m ² : 3.165	88.7	74.6				
According to geographic location/population background									
Barghi <i>et al.</i> , 2024 ⁷⁵	Meta-analysis	*	Asia: 5.39	87.72	92.11	<i>As quoted above</i>	*	*	
			Europe: 4.69	97.92	89.58				
			Africa: 5.93	100	87.5				
			United States: 5.88	90.00	100				

AMH, anti-Müllerian hormone; BMI, body mass index; ELFA, enzyme-linked fluorescent assay; ELISA, enzyme-linked immunosorbent assay; JSOG, Japan Society of Obstetrics and Gynaecology; PCOS, polycystic ovary syndrome; yo, years old.

*Data referring to a global search strategy with multiple AMH assays. Immunoassays are the preferred methods for AMH measurement.

^aIncludes the most representative data of the last 5 years.

^bExpressed as positive predictive value (PPV).

^cExpressed as negative predictive value (NPV).

a key factor influencing its levels.^{70,85} However, the observation of elevated AMH levels in women with normal ovulatory function, particularly over 40 years of age, raises critical concerns regarding the interpretation of AMH as a sole biomarker for PCOS in clinical settings.⁸⁸ As AMH is commonly associated with ovarian reserve and is frequently utilized in the evaluation of conditions like PCOS, the presence of elevated levels in otherwise healthy women complicate the clinical landscape.⁸⁹ This complexity highlights the need to gather the available evidence in a manner that could go beyond existing literature, incorporating a nuanced analysis of AMH values across diverse patient populations, including healthy women, older women, and those diagnosed with reproductive disorders. Future research should not only address the sensitivity and specificity of AMH as a diagnostic tool but also explore its positive and negative predictive values in these varied contexts. By synthesizing data from multiple studies that account for age, ethnicity, and reproductive health, researchers can establish more refined reference ranges and diagnostic criteria. This comprehensive approach will ultimately enhance the accuracy of PCOS diagnosis and improve reproductive health management strategies for women across different demographics.

The influence of insulin resistance (IR) and obesity on AMH

IR and obesity can influence AMH levels, particularly in women with PCOS.⁹⁰ Many studies indicate that IR can be associated with elevated AMH levels due to increased ovarian follicle counts, as reactive hyperinsulinemia stimulates ovarian theca cells to secrete androgens, which in turn impacts on AMH secretion.^{91,92} Similarly, obesity is often associated with IR and thus with high AMH levels via the same mechanisms.⁹³ However, the relationship between obesity and IR is not a linear, and in the other end of the spectrum, obesity can also be associated with disrupted ovarian function and low AMH levels.⁹⁴

Notwithstanding, elevated AMH levels in women with obesity and IR can complicate assessments of ovarian reserve and fertility potential, particularly in the context of PCOS, where metabolic health issues can further exacerbate reproductive challenges.⁹⁵ Moreover, intervention studies suggest that weight loss and lifestyle changes can improve insulin sensitivity and potentially affect AMH levels, but responses may vary among individuals.⁹⁶ Overall, understanding the interplay between obesity, IR, and AMH is crucial for addressing reproductive health and fertility in affected populations.

The impact of ethnic and geographic variability on AMH levels

Research indicates that ethnic and population background across different geographic regions significantly influence AMH levels, which can markedly differ among ethnic groups.⁹⁷ For instance, women of Asian descent often display lower AMH levels compared with Caucasian women, while those of African descent may present with higher levels. These differences can be attributed to a combination of genetic, lifestyle, and environmental factors such as diet and exposure to endocrine disruptors.⁹⁷ Additionally, variations in AMH levels are often observed in the context of PCOS, with certain populations, such as women from Middle Eastern or

Mediterranean regions, exhibiting higher AMH levels and a greater prevalence of this condition.⁹⁸

Geographic factors also play a role, as AMH levels can vary based on regional lifestyle, health practices, and socioeconomic conditions.⁹⁹ For example, women living in urban areas may show different AMH levels compared with those in rural settings due to variations in health-seeking behaviour and dietary habits.¹⁰⁰ Cultural practices related to fertility, such as trends toward delayed childbearing or specific dietary habits, can further affect reproductive hormone levels.¹⁰¹

The evidence available on the potential use of AMH for PCOS diagnosis predominantly stems from research studies conducted in Middle East and Asia,^{22,63,71,72,76-80} with a noticeable scarcity of studies from Europe and the United States.^{23,73-75} This disproportion raises important questions regarding the generalizability of AMH thresholds established in these populations to Western populations, particularly since AMH levels can be influenced by a range of ethnic, genetic, and environmental factors.¹⁰² Thus, the lack of robust data from diverse Western populations may limit the applicability of existing diagnostic criteria. It is imperative to understand whether the cut-offs identified in these studies are universally valid or reflect region-specific characteristics that may not be present in different demographic contexts. The absence of comprehensive studies in Western countries underlines the need for further investigation to establish population-specific AMH benchmarks, which would facilitate more accurate diagnosis and management of PCOS across various ethnic and geographic backgrounds.

Understanding these ethnic and geographic influences is crucial for accurate assessments of ovarian reserve and fertility potential, emphasizing the need for population-specific reference ranges in clinical practice. Acknowledging these differences can also guide personalized approaches in fertility treatments and interventions. Further research is essential to explore the underlying mechanisms behind these variations and to establish reliable AMH reference ranges across diverse populations.

AMH cut-off values for PCOS diagnosis

Considering the exponential growth in data regarding the use of AMH for PCOS diagnosis, research efforts have been directed, in part, to investigate AMH diagnostic cut-offs (Table 1, Figure 1).^{58,73,75,76,103-105} AMH levels between 3.5 ng/mL and 5 ng/mL have been frequently used as a cut-off, though other studies point at values above 5 ng/mL.^{58,59,72,103} Evidence has shown that using these cut-offs provide good sensitivity and specificity when used in conjunction with clinical features, such as HA and oligo-anovulation, but the use of a single cut-off value has been criticized due to age-related variations.^{17,74,78,106} A 2024 extended meta-analysis using a marginal beta-binomial model concluded that a serum AMH concentration of at least 5.39 ng/mL was associated with risk of PCOS (sensitivity, 88.6%; specificity, 92.75%).⁷⁵ Another recent systematic review and meta-analysis confirmed that the available data does not allow to define a universal cut-off value for PCOS diagnosis.¹⁷ In fact, the natural fluctuation in AMH levels along the lifespan, which peak by the age of 20 to 25 years, needs to be considered while analysing and defining cut-off values.⁷ Even though it has been widely demonstrated that AMH levels are useful for the diagnosis of PCOS in adolescents, regardless their corpulence,^{77,107} international guidelines still do not recommend the use of AMH as a diagnostic tool in this population and up to 8 years

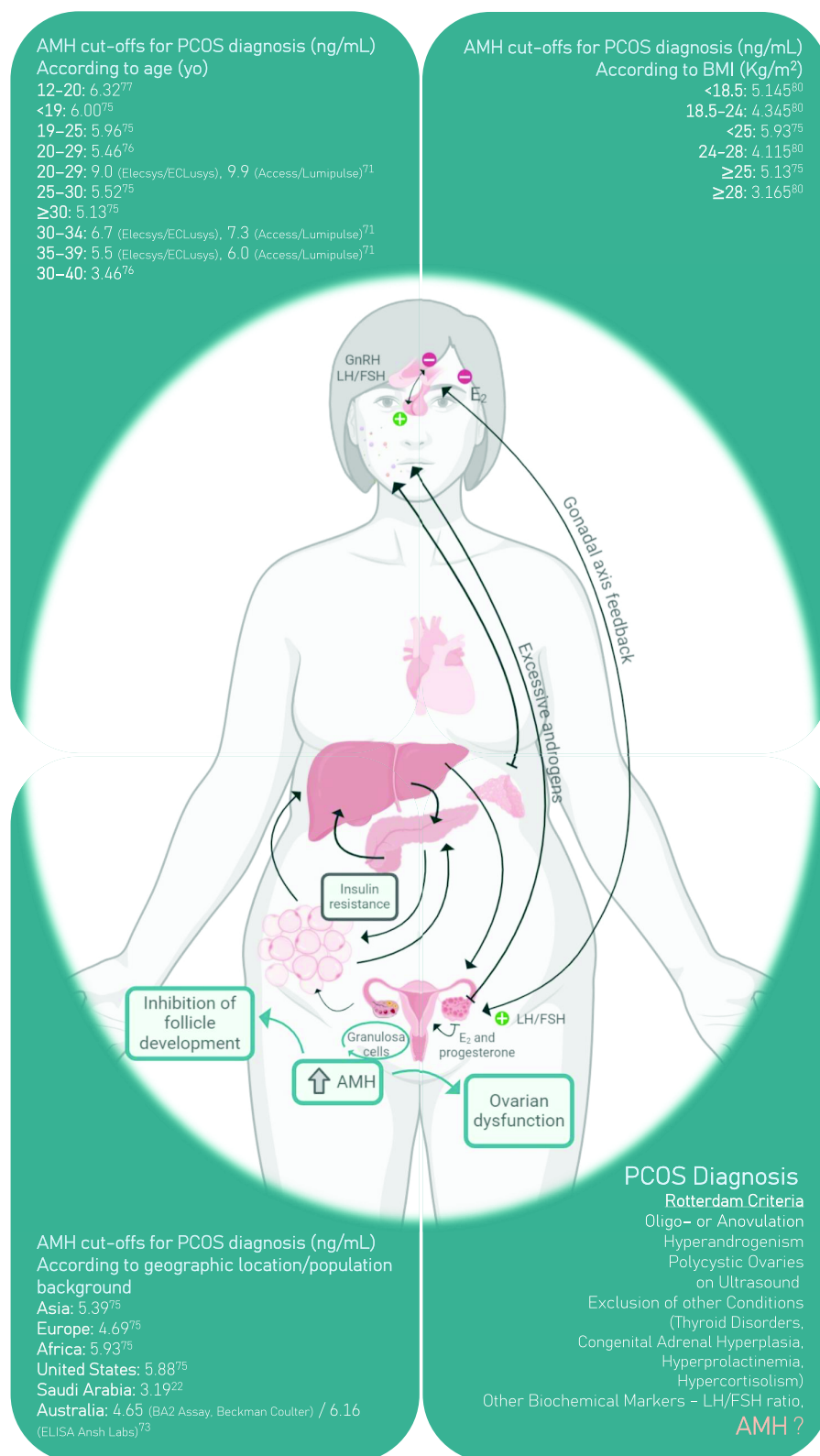


Figure 1. Diagram illustrating the different anti-Müllerian hormone (AMH) cut-offs proposed for polycystic ovary syndrome (PCOS) diagnosis, categorized according to age (in years old), body mass index (BMI), and geographic location/population background. E₂, estradiol; FSH, follicle-stimulating hormone; GnRH, gonadotropin-releasing hormone; LH, luteinizing hormone. [Created in BioRender. Vale-Fernandes, E. (2025) <https://BioRender.com/niykh8b>].

from menarche.^{7,108} However, research has shown that adjusting AMH levels for age and age-specific cut-offs improve the diagnostic power of AMH for PCOS.^{74,78,79,109,110}

In addition, AMH values can vary depending on the assay used,^{58,64,68,110} individual specific factors, like body weight,^{111,112} geographical and ethnic background.^{71,72,102}

As a matter of fact, in a 2019 review, Teede and co-workers⁵⁸ had already appointed the use of different AMH assays as a reason for the broad ranges of serum AMH values reported across studies and a major limitation for the establishment of cut-offs in the context of PCOS diagnosis. In recognition of this limitation, several authors focused on understanding how different assays for measuring AMH serum levels compared in an attempt of standardization. A comparative assessment of five different assays for the measurement of AMH serum levels showed an absence or weak bias between three different manual ELISA assays and evidenced lower AMH values for the automated assays.¹¹³ Overall, the authors concluded that upon receiver operating characteristic (ROC) analysis, the five assays showed similar ability to discriminate between women with and without PCOS, if different cut-off values were used for manual and automatic assays. A similar study which compared three assays, two manual and one automated, showed that the inter-assay correlation coefficient was AMH level-dependent,⁶⁸ again with the automated assay yielding lower levels as in the previously mentioned study.

Since it is evident that the discrepancies between assays can have major clinical implications, standardization is critical to determine the accuracy of AMH for PCOS diagnosis. For this to be achievable, the use of international reference preparations for precise calibration and standardized analyte-specific antibodies will be key to enhance comparability across assays.⁸¹ Additionally, implementing standard operating procedures in the pre-analytical phase is also important to minimize variability. These include the implementation of strict timing for blood sampling in order to reduce the impact of diurnal variation on AMH levels, which over 24 hours can range from 13% to 63% in women with PCOS and 10%-230% in healthy women.¹¹⁴ Additionally, it is important to take into consideration that some drugs, particularly GnRH (gonadotropin-releasing hormone) agonists, can also artificially elevate serum AMH levels.¹¹⁵ Awareness of these pre-analytical factors, alongside the implementation of standardized pre and analytical procedures, will be vital for the interpretation of assay results and enhance the reliability AMH use to support PCOS diagnosis and condition related clinical decisions.^{64,81,116,117}

Regarding ethnical and geographic variations, data show that in women from Saudi Arabia and African countries AMH cut-off values below 4.0 ng/mL had diagnostic power for PCOS,^{22,72} whereas higher AMH values (4.2 and 5.6 ng/mL) seem to be more predictive in European^{80,113} and even higher (4.4 ng/mL to 10 ng/mL) in Asian populations.^{63,77} A 2015 cross-sectional study showed that after controlling for potentially confounding factors, including age, BMI, and smoking status, the differences in AMH levels between ethnical groups were no longer significant.¹¹⁸ However, a recent meta-analysis presented a subgroup analysis, including the available data until August 2024, of AMH cut-off values for the diagnosis of PCOS and retrieved higher values for African (5.93 ng/mL) and United States (5.88 ng/mL).⁷⁵

Focusing on individual particularities and comorbidities associated with PCOS, namely the impact of obesity and IR, which are prevalent in women with PCOS, on the diagnostic power of AMH has been under discussion.^{24,119,120} Even though, the data on the impact of obesity, defined by BMI values, on AMH levels are conflicting, probably due to the small number of individuals in some cohorts, most studies tend to show a negative relationship between AMH levels and BMI.¹²¹ Of notice, a recent study, which included 3160 participants showed that a greater

BMI seems to be associated with a lower AMH cut-off for PCOS diagnosis. The researchers developed a model, combining AMH and BMI, to calculate the specific probabilities of diagnosis.¹²² In contrast, although several studies suggested that there could be a relationship between AMH levels and IR, a recent systematic review and meta-analysis, which included 22 articles, concluded that there was no evidence of a significant relationship between AMH levels and IR in women with PCOS.¹²³

Overall, the study of the correlations between AMH levels and different PCOS phenotypes has evidenced not only a correlation between AMH levels and PCOS severity but also with other clinical and biochemical particularities, while supporting the definition of AMH diagnostic cut-offs.⁹⁸ As such, AMH seems to be of value to distinguish between different PCOS phenotypes, further contributing to refine diagnosis and consider personalized treatment approaches.^{33,36,119}

Regarding PCOS phenotypes, Li and co-workers¹²⁴ proposed that AMH, using a cut-off value of 4.23 ng/mL, was only adequate to predict PCOS in women with HA with a sensitivity of 82%. However, other studies have suggested a limited correlation between androgens and AMH levels.^{62,125} A recent study including different clinical and biochemical characteristics, such as age, body weight, and BMI, proposed that AMH-based phenotyping seems to be a cost-effective tool to explore the PCOS heterogeneity.³³ However, the discussion of the impact of AMH levels on PCOS phenotyping is not free from controversy, with data showing heterogeneity on the association between high AMH levels, OD, and HA in the different PCOS phenotypes.¹²⁶

Pros and cons of AMH for PCOS diagnosis

AMH offers several advantages in PCOS diagnosis. It is a non-invasive and objective biomarker with high sensitivity in PCOM. Non-invasiveness is particularly relevant for adolescents, as it allows to avoid intravaginal ultrasound. Its direct correlation with follicular count and small fluctuations throughout the menstrual cycle make it a more consistent marker when compared to hormonal assays like LH (luteinizing hormone)/FSH ratios. Additionally, it reduces interobserver variability linked to ultrasound-based antral follicle counts, which can be associated with reduced accuracy, particularly in women with obesity. AMH measurements may also improve diagnosis in women with irregular cycles, aiding clinicians in timely interventions.

However, the use of AMH for diagnostic purposes is not without drawbacks. Elevated AMH levels are not exclusive to PCOS and can occur in other ovarian pathologies, such as hypogonadotropic hypogonadism, decreasing its diagnostic specificity.¹²⁷ Moreover, AMH assays lack standardization, which leads to variability in results across methodologies^{64,68,81}; the lack of standardized cut-off values is also a major challenge that needs to be addressed, eventually considering regional and ethnical variability. Concerns also remain regarding the interpretation of AMH levels in adolescents and aging women, where natural fluctuations may overlap with PCOS diagnostic cut-offs.^{74,78} In addition, the effect of overweight on AMH levels may also have to be considered.¹²² Overall, the use of AMH as a single marker for PCOS has been shown to have poor sensitivity and specificity and, as such, there is no evidence for it to be recommended.^{7,17,60,128}

Emerging consensus suggests that while AMH is a valuable biomarker for diagnosing PCOS, it should not replace the

current criteria but rather be integrated as additional biomarker.⁷ Researchers continue to call for global standardization of AMH assays and population-specific cut-offs definitions to enhance its diagnostic utility. Future research should focus on exploring the use of AMH in different prediction models towards precision diagnosis of PCOS across its phenotypic diversity. This thorough diagnostic process is crucial for ensuring accurate diagnosis and facilitating appropriate clinical interventions for women with PCOS. In summary, while AMH could enhance the diagnostic accuracy of the Rotterdam Criteria, how should this be implemented into the current criteria and to what extent will this improve diagnosis across different PCOS phenotypes, still requires to be demonstrated.

Conclusions

In conclusion, while AMH is promising as a biomarker for fine-tuning precision diagnosis of PCOS, its incorporation into current diagnostic criteria remains complex. AMH offers advantages such as its non-invasive nature, high sensitivity to PCOM, and consistency across the menstrual cycle. However, challenges persist due to variability in AMH assay methods, lack of standardized cut-off values, and the influence of factors like age and obesity on AMH levels. The predominance of studies from Middle East and Asia, with a notable lack of data from Europe and the USA, may highlight the disparities in access to ultrasound diagnostics, suggesting a greater unmet need for reliable biochemical markers in regions with less diagnostic availability. These challenges undermine the promising potential of AMH as a reliable biomarker for refining diagnostic criteria in PCOS. The exploration of AMH in various populations, standardization of measurement techniques, and comprehensive studies correlating AMH levels with clinical outcomes in PCOS could further elucidate its role as a biomarker. By providing insight into ovarian reserve and activity, AMH may enhance the specificity and sensitivity of PCOS diagnoses, particularly in cases where symptoms are ambiguous or overlap with other conditions. Incorporating AMH levels into the diagnostic framework could lead to more accurate and personalized treatment strategies, ultimately improving reproductive health outcomes for women experiencing fertility issues related to PCOS. To fully harness the potential of AMH as a reliable biomarker for PCOS diagnosis, it is essential to conduct future prospective multicentred studies in order to establish standardized cut-off values incorporating relevant co-factors for interpretation and aligned with commercial assay methodologies.

Acknowledgments

The authors would like to thank to Paula Pinto, PharmD, PhD (PMA – Pharmaceutical Medicine Academy) for providing medical writing and editorial assistance.

Funding

UMIB-Unit for Multidisciplinary Research in Biomedicine is funded by Foundation for Science and Technology (FCT) through institutional grants (UIDB/00215/2020 and UIDP/00215/2020); ITR-Laboratory for Integrative and Translational Research in Population Health is funded by FCT through institutional grants (LA/P/0064/2020). The professional editing of this manuscript was supported by an unrestricted grant from Organon Portugal, Sociedade Unipessoal, LDA.

Authors' contributions

Emídio Vale-Fernandes (Conceptualization [equal], Data curation [equal], Formal analysis [equal], Funding acquisition [equal], Investigation [equal], Methodology [equal], Resources [equal], Writing—original draft [equal], Writing—review & editing [equal]), Duarte Pignatelli (Validation [equal], Visualization [equal], Writing—review & editing [equal]), and Mariana Monteiro (Supervision [equal], Validation [equal], Visualization [equal], Writing—review & editing [equal])

E.V.-F., D.P. and M.P.M. performed literature review, interpreted the data, and wrote the article. All authors contributed to revising the article and approved the final version of the manuscript to be published.

Conflict of interest: None declared. The financial support given by Organon Portugal, Sociedade Unipessoal, LDA, for editing the manuscript had no impact on the research.

Data availability

No new data were generated or analysed in support of this research.

References

1. Organization, W.H. *Polycystic ovary syndrome*. 2023; World Health Organization. Accessed January 17, 2025. <https://www.who.int/news-room/fact-sheets/detail/polycystic-ovary-syndrome>
2. Christ JP, Cedars MI. Current guidelines for diagnosing PCOS. *Diagnostics*. 2023;13(6):1113. <https://doi.org/10.3390/diagnostics13061113>
3. Rotterdam ESHRE/ASRM-Sponsored PCOS consensus workshop group. Revised 2003 consensus on diagnostic criteria and long-term health risks related to polycystic ovary syndrome (PCOS). *Hum Reprod*. 2004;19(1):41-47. <https://doi.org/10.1093/humrep/deh098>
4. Asanidze E, Kristesashvili J, Parunashvili N, Karelshvili N, Etsadashvili N. Challenges in diagnosis of polycystic ovary syndrome in adolescence. *Gynecol Endocrinol*. 2021;37(9):819-822. <https://doi.org/10.1080/09513590.2021.1943344>
5. Kim JJ, Hwang KR, Lee D, Kim S, Choi YM. Adolescents diagnosed with polycystic ovary syndrome under the rotterdam criteria but not meeting the diagnosis under the updated guideline. *Hum Reprod*. 2024;39(5):1072-1077. <https://doi.org/10.1093/humrep/deae042>
6. Teede HJ, Tay CT, Laven JJE, *et al*. Recommendations from the international evidence-based guideline for the assessment and management of polycystic ovary syndrome. *Hum Reprod*. 2018;33(9):1602-1618. <https://doi.org/10.1093/humrep/dey256>
7. Teede HJ, Tay CT, Laven JJE, *et al*. Recommendations from the 2023 international evidence-based guideline for the assessment and management of polycystic ovary syndrome. *J Clin Endocrinol Metab*. 2023;108(10):2447-2469. <https://doi.org/10.1210/clinem/dgad463>
8. Fahs D, Salloom D, Nasrallah M, Ghazeeri G. Polycystic ovary syndrome: pathophysiology and controversies in diagnosis. *Diagnostics (Basel)*. 2023;13(9):1559. <https://doi.org/10.3390/diagnostics13091559>
9. Beil K, Dewailly D, Seemann R, *et al*. Polycystic ovary syndrome phenotype D versus functional hypothalamic amenorrhea with polycystic ovarian morphology: a retrospective study about a frequent differential diagnosis. *Front Endocrinol (Lausanne)*. 2022;13:904706. <https://doi.org/10.3389/fendo.2022.904706>
10. Dewailly D. Diagnostic criteria for PCOS: is there a need for a rethink? *Best Pract Res Clin Obstet Gynaecol*. 2016;37:5-11. <https://doi.org/10.1016/j.bpobgyn.2016.03.009>

11. Bellver J, Rodríguez-Tabernero L, Robles A, *et al.* Polycystic ovary syndrome throughout a woman's life. *J Assist Reprod Genet.* 2018;35(1):25-39. <https://doi.org/10.1007/s10815-017-1047-7>
12. Goodman NF, Cobin RH, Futterweit W, *et al.* AMERICAN ASSOCIATION OF CLINICAL ENDOCRINOLOGISTS, AMERICAN COLLEGE OF ENDOCRINOLOGY, AND ANDROGEN EXCESS AND PCOS SOCIETY DISEASE STATE CLINICAL REVIEW: GUIDE TO THE BEST PRACTICES IN THE EVALUATION AND TREATMENT OF POLYCYSTIC OVARY SYNDROME--PART 1. *Endocr Pract.* 2015;21(11):1291-1300. <https://doi.org/10.4158/EP15748.DSC>
13. Lauritsen MP, Pinborg A, Loft A, *et al.* Revised criteria for PCOS in WHO group II anovulatory infertility - a revival of hypothalamic amenorrhoea? *Clin Endocrinol (Oxf).* 2015;82(4):584-591. <https://doi.org/10.1111/cen.12621>
14. Huffman AM, Reza S, Basnet J, Romero DG. Biomarkers in polycystic ovary syndrome. *Curr Opin Physiol.* 2023;36:100717. <https://doi.org/10.1016/j.cophys.2023.100717>
15. Carmina E, Lobo RA. Comparing lean and obese PCOS in different PCOS phenotypes: evidence that the body weight is more important than the Rotterdam phenotype in influencing the metabolic status. *Diagnostics (Basel).* 2022;12(10):2313. <https://doi.org/10.3390/diagnostics12102313>
16. Sivanandy MS, Ha SK. The role of serum anti-Müllerian hormone measurement in the diagnosis of polycystic ovary syndrome. *Diagnostics (Basel).* 2023;13(5):907. <https://doi.org/10.3390/diagnostics13050907>
17. van der Ham K, Laven JSE, Tay CT, Mousa A, Teede H, Louwers YV. Anti-Müllerian hormone as a diagnostic biomarker for polycystic ovary syndrome and polycystic ovarian morphology: a systematic review and meta-analysis. *Fertil Steril.* 2024;122(4):727-739. <https://doi.org/10.1016/j.fertnstert.2024.05.163>
18. La Marca A, Broekmans FJ, Volpe A, *et al.* Anti-Müllerian hormone (AMH): what do we still need to know? *Hum Reprod.* 2009;24(9):2264-2275. <https://doi.org/10.1093/humrep/dep210>
19. Hart R, Doherty DA, Norman RJ, *et al.* Serum antimüllerian hormone (AMH) levels are elevated in adolescent girls with polycystic ovaries and the polycystic ovarian syndrome (PCOS). *Fertil Steril.* 2010;94(3):1118-1121. <https://doi.org/10.1016/j.fertnstert.2009.11.002>
20. Lin YH, Chiu WC, Wu CH, Tzeng CR, Hsu CS, Hsu MI. Antimüllerian hormone and polycystic ovary syndrome. *Fertil Steril.* 2011;96(1):230-235. <https://doi.org/10.1016/j.fertnstert.2011.04.003>
21. Piltonen TT, Komi E, Morin-Papunen LC, *et al.* AMH as part of the diagnostic PCOS workup in large epidemiological studies. *Eur J Endocrinol.* 2023;188(6):547-554. <https://doi.org/10.1093/ejendo/ivad065>
22. Ahmed N, Batarfi AA, Bajouh OS, Bakhshab S. Serum Anti-Müllerian hormone in the diagnosis of polycystic ovary syndrome in association with clinical symptoms. *Diagnostics (Basel).* 2019;9(4):136. <https://doi.org/10.3390/diagnostics9040136>
23. Anand S, Kumar A, Prasad A, Trivedi K. Updated meta-analysis on the diagnostic accuracy of serum anti-Müllerian hormone in polycystic ovary syndrome involving 13 509 subjects. *J Obstet Gynaecol Res.* 2022;48(8):2162-2174. <https://doi.org/10.1111/jog.15233>
24. Jun TJ, Jelani AM, Omar J, Rahim RA, Yaacob NM. Serum Anti-Müllerian hormone in polycystic ovary syndrome and its relationship with insulin resistance, lipid profile and adiponectin. *Indian J Endocrinol Metab.* 2020;24(2):191-195. https://doi.org/10.4103/ijem.IJEM_305_19
25. Chang S, Dunaif A. Diagnosis of polycystic ovary syndrome: which criteria to use and when? *Endocrinol Metab Clin North Am.* 2021;50(1):11-23. <https://doi.org/10.1016/j.ecl.2020.10.002>
26. Stein IF, Leventhal ML. Amenorrhea associated with bilateral polycystic ovaries. *Am J Obstet Gynecol.* 1935;29(2):181-191. [https://doi.org/10.1016/S0002-9378\(15\)30642-6](https://doi.org/10.1016/S0002-9378(15)30642-6)
27. Zawadzki JA, Dunaif A. Diagnostic criteria for polycystic ovary syndrome: towards a rational approach. In: Dunaif A, Givens JR, Haseltine F, eds. *Polycystic Ovary Syndrome*. Vol 10. Boston, MA, USA: Blackwell Scientific Publications Cambridge; 1992:377-384.
28. Azziz R, Carmina E, Dewailly D, *et al.* The androgen excess and PCOS society criteria for the polycystic ovary syndrome: the complete task force report. *Fertil Steril.* 2009;91(2):456-488. <https://doi.org/10.1016/j.fertnstert.2008.06.035>
29. Azziz R, Carmina E, Dewailly D, *et al.* Positions statement: criteria for defining polycystic ovary syndrome as a predominantly hyperandrogenic syndrome: an androgen excess society guideline. *J Clin Endocrinol Metab.* 2006;91(11):4237-4245. <https://doi.org/10.1210/jc.2006-0178>
30. Gruiters MJ, Visser JA, Durlinger AL, Themmen AP. Anti-Müllerian hormone and its role in ovarian function. *Mol Cell Endocrinol.* 2003;211(1-2):85-90. <https://doi.org/10.1016/j.mce.2003.09.024>
31. Pigny P, Jonard S, Robert Y, Dewailly D. Serum Anti-Müllerian hormone as a surrogate for antral follicle count for definition of the polycystic ovary syndrome. *J Clin Endocrinol Metab.* 2006;91(3):941-945. <https://doi.org/10.1210/jc.2005-2076>
32. Pellatt L, Rice S, Dilaver N, *et al.* Anti-Müllerian hormone reduces follicle sensitivity to follicle-stimulating hormone in human granulosa cells. *Fertil Steril.* 2011;96(5):1246-51.e1. <https://doi.org/10.1016/j.fertnstert.2011.08.015>
33. Nguyen MT, Krishnan S, Phatak SV, Karakas SE. Anti-Müllerian hormone-based phenotyping identifies subgroups of women with polycystic ovary syndrome with differing clinical and biochemical characteristics. *Diagnostics (Basel).* 2023;13(3):500. <https://doi.org/10.3390/diagnostics13030500>
34. Woo HY, Kim KH, Rhee EJ, Park H, Lee MK. Differences of the association of anti-Müllerian hormone with clinical or biochemical characteristics between women with and without polycystic ovary syndrome. *Endocr J.* 2012;59(9):781-790. <https://doi.org/10.1507/endocrj.EJ12-0055>
35. Jabarpour M, Amidi F, Rostami S, *et al.* High Anti-Müllerian hormone strongly correlates with reproductive outcomes in women undergoing assisted reproduction. *Clin Lab.* 2023; 69(12):2479-2483. <https://doi.org/10.7754/Clin.Lab.2023.230524>
36. Liu S, Hong L, Mo M, *et al.* Association of antimüllerian hormone with polycystic ovarian syndrome phenotypes and pregnancy outcomes of in vitro fertilization cycles with fresh embryo transfer. *BMC Pregnancy Childbirth.* 2022;22(1):171. <https://doi.org/10.1186/s12884-022-04518-0>
37. Hou Y, Wang L, Li Y, Ai J, Tian L. Serum levels of anti-Müllerian hormone influence pregnancy outcomes associated with gonadotropin-releasing hormone antagonist treatment: a retrospective cohort study. *Sci Rep.* 2023;13(1):2127. <https://doi.org/10.1038/s41598-023-28724-8>
38. Dumont A, Robin G, Catteau-Jonard S, Dewailly D. Role of anti-Müllerian hormone in pathophysiology, diagnosis and treatment of polycystic ovary syndrome: a review. *Reprod Biol Endocrinol.* 2015;13(1):137. <https://doi.org/10.1186/s12958-015-0134-9>
39. Rudnicka E, Kunicki M, Calik-Ksepka A, *et al.* Anti-Müllerian hormone in pathogenesis, diagnostic and treatment of PCOS. *Int J Mol Sci.* 2021;22(22):12507. <https://doi.org/10.3390/ijms222212507>
40. Chen X, Hong L, Diao L, Yin T, Liu S. Hyperandrogenic environment regulates the function of ovarian granulosa cells by modulating macrophage polarization in PCOS. *Am J Reprod Immunol.* 2024;91(5):e13854. <https://doi.org/10.1111/aji.13854>
41. Vale-Fernandes E, Moreira MV, Rodrigues B, *et al.* Anti-Müllerian hormone a surrogate of follicular fluid oxidative stress in polycystic ovary syndrome? *Front Cell Dev Biol.* 2024;12:1408879. <https://doi.org/10.3389/fcell.2024.1408879>
42. Bedenk J, Vrtačnik-Bokal E, Virant-Klun I. The role of anti-Müllerian hormone (AMH) in ovarian disease and infertility.

- J Assist Reprod Genet.* 2020;37(1):89-100. <https://doi.org/10.1007/s10815-019-01622-7>
43. Moreira MV, Vale-Fernandes E, Albergaria IC, Alves MG, Monteiro MP. Follicular fluid composition and reproductive outcomes of women with polycystic ovary syndrome undergoing in vitro fertilization: a systematic review. *Rev Endocr Metab Disord.* 2023;24(6):1045-1073. <https://doi.org/10.1007/s11154-023-09819-z>
 44. Tal R, Tal O, Seifer BJ, Seifer DB. Antimüllerian hormone as predictor of implantation and clinical pregnancy after assisted conception: a systematic review and meta-analysis. *Fertil Steril.* 2015;103(1):119-130.e3. <https://doi.org/10.1016/j.fertnstert.2014.09.041>
 45. La Marca A, Ferraretti AP, Palermo R, Ubaldi FM. The use of ovarian reserve markers in IVF clinical practice: a national consensus. *Gynecol Endocrinol.* 2016;32(1):1-5. <https://doi.org/10.3109/09513590.2015.1102879>
 46. Poseidon Group (Patient-Oriented Strategies Encompassing Individualized Oocyte Number); Alviggi C, Andersen CY, *et al.* A new more detailed stratification of low responders to ovarian stimulation: from a poor ovarian response to a low prognosis concept. *Fertil Steril.* 2016;105(6):1452-1453. <https://doi.org/10.1016/j.fertnstert.2016.02.005>
 47. Yuwen T, Yang Z, Cai G, Feng G, Liu Q, Fu H. Association between serum AMH levels and IVF/ICSI outcomes in patients with polycystic ovary syndrome: a systematic review and meta-analysis. *Reprod Biol Endocrinol.* 2023;21(1):95. <https://doi.org/10.1186/s12958-023-01153-y>
 48. Reis AB, Leal C, Barreiro M, Tomé A, Vale-Fernandes E. Anti-Müllerian hormone as a predictor of the number of oocytes obtained during in vitro fertilization treatments. *JBRA Assist Reprod.* 2024;28(4):597-603. <https://doi.org/10.5935/1518-0557.20240049>
 49. Li HWR, Nelson SM. Clinical application of AMH measurement in assisted reproduction. *Front Endocrinol (Lausanne).* 2020;11:606744. <https://doi.org/10.3389/fendo.2020.606744>
 50. Guo Y, Jiang H, Hu S, Liu S, Li F, Jin L. Efficacy of three COS protocols and predictability of AMH and AFC in women with discordant ovarian reserve markers: a retrospective study on 19,239 patients. *J Ovarian Res.* 2021;14(1):111. <https://doi.org/10.1186/s13048-021-00863-4>
 51. Vale-Fernandes E, Barreiro M, Monteiro MP. Candidates selection for oocyte donation in a public gamete bank—predictive value of the anti-Müllerian hormone. *Porto Biomed J.* 2023;8(1):e199. <https://doi.org/10.1097/j.pbj.0000000000000199>
 52. Oliveira T, Barreiro M, Tomé A, Vale-Fernandes E. Female reproductive health – study of an egg donor population. *JBRA Assist Reprod.* 2023;27(4):638-643. <https://doi.org/10.5935/1518-0557.20230046>
 53. Tal R, Seifer CM, Khanimov M, Seifer DB, Tal O. High serum antimüllerian hormone levels are associated with lower live birth rates in women with polycystic ovarian syndrome undergoing assisted reproductive technology. *Reprod Biol Endocrinol.* 2020;18(1):20. <https://doi.org/10.1186/s12958-020-00581-4>
 54. Guan Y, Kong P, Xiao Z, *et al.* Independent variables for determining the cumulative live birth rates of aged patients with polycystic ovary syndrome or tubal factor infertility: a retrospective cohort study. *Front Endocrinol (Lausanne).* 2021;12:728051. <https://doi.org/10.3389/fendo.2021.728051>
 55. Vale-Fernandes E, Barreiro M, Leal C, Macedo RZ, Tomé A, Monteiro MP. Elevated anti-Müllerian hormone as a prognostic factor for poor outcomes of in vitro fertilization in women with polycystic ovary syndrome. *Biomedicine.* 2023;11(12):3150. <https://doi.org/10.3390/biomedicine11123150>
 56. Xu HY, Zhang HX, Xiao Z, Qiao J, Li R. Regulation of anti-Müllerian hormone (AMH) in males and the associations of serum AMH with the disorders of male fertility. *Asian J Androl.* 2019;21(2):109-114. https://doi.org/10.4103/aja.aja_83_18
 57. Zeng Y, Zhao G, Zheng Y, Jiang X. From biological marker to clinical application: the role of anti-Müllerian hormone for delayed puberty and idiopathic non-obstructive azoospermia in males. *Endocr Connect.* 2025;14(3):e240630. <https://doi.org/10.1530/EC-24-0630>
 58. Teede H, Misso M, Tassone EC, *et al.* Anti-Müllerian hormone in PCOS: a review informing international guidelines. *Trends Endocrinol Metab.* 2019;30(7):467-478. <https://doi.org/10.1016/j.tem.2019.04.006>
 59. Chaabouni K, Makhoul R, Khanfir F, *et al.* The anti-Müllerian hormone: a useful tool for diagnosis of polycystic ovary syndrome. *Ann Biol Clin (Paris).* 2023;81(4):410-416. <https://doi.org/10.1684/abc.2023.1830>
 60. Piltonen TT, Viita-Aho J, Saarela U, Melin J, Forslund M. Utility of serum anti-Müllerian hormone measurement as part of polycystic ovary syndrome diagnosis. *Semin Reprod Med.* 2024;42((011)):49-59. <https://doi.org/10.1055/s-0044-1786731>
 61. Iliodromiti S, Kelsey TW, Anderson RA, Nelson SM. Can anti-Müllerian hormone predict the diagnosis of polycystic ovary syndrome? A systematic review and meta-analysis of extracted data. *J Clin Endocrinol Metab.* 2013;98(8):3332-3340. <https://doi.org/10.1210/jc.2013-1393>
 62. Sahmay S, Atakul N, Aydogan B, Aydin Y, Imamoglu M, Seyisoglu H. Elevated serum levels of anti-Müllerian hormone can be introduced as a new diagnostic marker for polycystic ovary syndrome. *Acta Obstet Gynecol Scand.* 2013;92(12):1369-1374. <https://doi.org/10.1111/aogs.12247>
 63. Wang L, Luo M, Yu X, *et al.* Assessing the clinical diagnostic value of anti-Müllerian hormone in polycystic ovarian syndrome and its correlation with clinical and metabolism indicators. *J Ovarian Res.* 2024;17(1):78. <https://doi.org/10.1186/s13048-024-01405-4>
 64. Punchoo R, Bhoora S. Variation in the measurement of Anti-Müllerian hormone - what are the laboratory issues? *Front Endocrinol (Lausanne).* 2021;12:719029. <https://doi.org/10.3389/fendo.2021.719029>
 65. Thaker N, Dhande R, Parihar P. Role of transvaginal sonography in the diagnosis of female infertility: a comprehensive review. *Cureus.* 2023;15(12):e50048. <https://doi.org/10.7759/cureus.50048>
 66. Pache TD, Hop WC, Wladimiroff JW, Schipper J, Fauser BC. Transvaginal sonography and abnormal ovarian appearance in menstrual cycle disturbances. *Ultrasound Med Biol.* 1991;17(6):589-593. [https://doi.org/10.1016/0301-5629\(91\)90029-V](https://doi.org/10.1016/0301-5629(91)90029-V)
 67. Sahmay S, Aydin Y, Oncul M, Senturk LM. Diagnosis of polycystic ovary syndrome: AMH in combination with clinical symptoms. *J Assist Reprod Genet.* 2014;31(2):213-220. <https://doi.org/10.1007/s10815-013-0149-0>
 68. Moolhuijsen LME, Louwers YV, Laven JSE, Visser JA. Comparison of 3 different AMH assays with AMH levels and follicle count in women with polycystic ovary syndrome. *J Clin Endocrinol Metab.* 2022;107(9):e3714-e3722. <https://doi.org/10.1210/clinem/dgac370>
 69. Pankhurst MW, Chong YH, McLennan IS. Enzyme-linked immunosorbent assay measurements of antimüllerian hormone (AMH) in human blood are a composite of the uncleaved and bioactive cleaved forms of AMH. *Fertil Steril.* 2014;101(3):846-850. <https://doi.org/10.1016/j.fertnstert.2013.12.009>
 70. Moolhuijsen LME, Visser JA. Anti-Müllerian hormone and ovarian reserve: update on assessing ovarian function. *J Clin Endocrinol Metab.* 2020;105(11):3361-3373. <https://doi.org/10.1210/clinem/dgaa513>
 71. Noguchi H, Iwasa T, Iwase A, *et al.* Cut-off value for anti-Müllerian hormone in the diagnostic criteria for polycystic ovary syndrome in the Japanese population. *J Obstet Gynaecol Res.* 2024;50(8):1368-1382. <https://doi.org/10.1111/jog.15972>
 72. Alhassan S, Elmugadam A, Elfadil GA, *et al.* Diagnostic performance of anti-Müllerian hormone, luteinizing hormone to follicle-stimulating hormone ratio, testosterone, and prolactin to predict polycystic ovary syndrome among Sudanese women. *Int J Womens Health.* 2023;15:837-843. <https://doi.org/10.2147/IJWH.S403347>

73. Bell RJ, Islam RM, Skiba MA, Herbert D, Martinez Garcia A, Davis SR. Substituting serum anti-Müllerian hormone for polycystic ovary morphology increases the number of women diagnosed with polycystic ovary syndrome: a community-based cross-sectional study. *Hum Reprod.* 2021;37(1):109-118. <https://doi.org/10.1093/humrep/deab232>
74. Evliyaoglu O, Imöhl M, Weiskirchen R, van Helden J. Age-specific reference values improve the diagnostic performance of AMH in polycystic ovary syndrome. *Clin Chem Lab Med.* 2020;58(8):1291-1301. <https://doi.org/10.1515/cclm-2019-1059>
75. Barghi M, Heidari Z, Haghighatdoost F, Feizi A, Hashemipour M. New insights into the relationship of antimüllerian hormone with polycystic ovary syndrome and its diagnostic accuracy: an updated and extended meta-analysis using a marginal beta-binomial model. *Am J Obstet Gynecol.* 2025;232(2):164-187.e31. <https://doi.org/10.1016/j.ajog.2024.10.004>
76. Kakkad V, Reddy NS, Nihlani H, Gundewar T. Age-related diagnostic threshold of anti-Müllerian hormone for polycystic ovarian syndrome. *Int J Gynaecol Obstet.* 2021;153(3):443-448. <https://doi.org/10.1002/ijgo.13515>
77. Li M, Ruan X, Ju R, et al. Is anti-Müllerian hormone a useful biomarker in the diagnosis of polycystic ovary syndrome in Chinese adolescents? *Gynecol Endocrinol.* 2022;38(2):148-152. <https://doi.org/10.1080/09513590.2021.2016694>
78. Ramezani Tehrani F, Rahmati M, Mahboobifard F, Firouzi F, Hashemi N, Azizi F. Age-specific cut-off levels of anti-Müllerian hormone can be used as diagnostic markers for polycystic ovary syndrome. *Reprod Biol Endocrinol.* 2021;19(1):76. <https://doi.org/10.1186/s12958-021-00755-8>
79. Timur HT, Cimrin D, Gursay Doruk O, Dogan OE. Determining the age group-based cut-off values of serum anti-Müllerian hormone concentrations to diagnose polycystic ovary syndrome. *Curr Med Res Opin.* 2023;39(6):855-863. <https://doi.org/10.1080/03007995.2023.2204768>
80. Zhang M, Liu X, Xu X, et al. The reference value of anti-Müllerian hormone to diagnose polycystic ovary syndrome is inversely associated with BMI: a retrospective study. *Reprod Biol Endocrinol.* 2023;21(1):15. <https://doi.org/10.1186/s12958-023-01064-y>
81. Demirci G, Saçintı KG, Sönmezer M, Öztürk HS. AMH standardization and preanalytical phase as two sides of the same coin. *JBRA Assist Reprod.* 2023;27(2):337-338. <https://doi.org/10.5935/1518-0557.20220076>
82. Hussain M, Cahill D, Akande V, Gordon U. Discrepancies between antimüllerian hormone and follicle stimulating hormone in assisted reproduction. *Obstet Gynecol Int.* 2013;2013:383278. <https://doi.org/10.1155/2013/383278>
83. Su HI, Sammel MD, Homer MV, Bui K, Haunschild C, Stanczyk FZ. Comparability of antimüllerian hormone levels among commercially available immunoassays. *Fertil Steril.* 2014;101(6):1766-72.e1. <https://doi.org/10.1016/j.fertnstert.2014.02.046>
84. Sinha S, Sharan A, Sinha S. Anti-Müllerian hormone as a marker of ovarian reserve and function. *Cureus.* 2022;14(9):e29214. <https://doi.org/10.7759/cureus.29214>
85. Cedars MI. Evaluation of female fertility-AMH and ovarian reserve testing. *J Clin Endocrinol Metab.* 2022;107(6):1510-1519. <https://doi.org/10.1210/clinem/dgac039>
86. Iwase A, Hasegawa Y, Tsukui Y, et al. Anti-Müllerian hormone beyond an ovarian reserve marker: the relationship with the physiology and pathology in the life-long follicle development. *Front Endocrinol (Lausanne).* 2023;14:1273966. <https://doi.org/10.3389/fendo.2023.1273966>
87. Freeman EW, Sammel MD, Lin H, Gracia CR. Anti-müllerian hormone as a predictor of time to menopause in late reproductive age women. *J Clin Endocrinol Metab.* 2012;97(5):1673-1680. <https://doi.org/10.1210/jc.2011-3032>
88. Tokura Y, Yoshino O, Ogura-Nose S, et al. The significance of serum anti-Müllerian hormone (AMH) levels in patients over age 40 in first IVF treatment. *J Assist Reprod Genet.* 2013;30(6):821-825. <https://doi.org/10.1007/s10815-013-9991-3>
89. Safer LZ, Grossman LC, Chan CW, Sauer MV, Lobo RA, Douglas NC. Use of anti-Müllerian hormone testing during ovarian reserve screening to identify women at risk of polycystic ovary syndrome. *Int J Gynaecol Obstet.* 2016;135(1):73-76. <https://doi.org/10.1016/j.ijgo.2016.03.026>
90. Zhao H, Zhou D, Liu C, Zhang L. The relationship between insulin resistance and obesity and serum anti-Müllerian hormone level in Chinese women with polycystic ovary syndrome: a retrospective, single-center cohort study. *Int J Womens Health.* 2023;15:151-166. <https://doi.org/10.2147/IJWH.S393594>
91. Nardo LG, Yates AP, Roberts SA, Pemberton P, Laing I. The relationships between AMH, androgens, insulin resistance and basal ovarian follicular status in non-obese subfertile women with and without polycystic ovary syndrome. *Hum Reprod.* 2009;24(11):2917-2923. <https://doi.org/10.1093/humrep/dep225>
92. Rosenfield RL, Ehrmann DA. The pathogenesis of polycystic ovary syndrome (PCOS): the hypothesis of PCOS as functional ovarian hyperandrogenism revisited. *Endocr Rev.* 2016;37(5):467-520. <https://doi.org/10.1210/er.2015-1104>
93. Zeng X, Huang Y, Zhang M, et al. Anti-Müllerian hormone was independently associated with central obesity but not with general obesity in women with PCOS. *Endocr Connect.* 2022;11(1):e210243. <https://doi.org/10.1530/EC-21-0243>
94. Bernardi LA, Carnethon MR, de Chavez PJ, et al. Relationship between obesity and anti-Müllerian hormone in reproductive-aged African American women. *Obesity (Silver Spring).* 2017;25(1):229-235. <https://doi.org/10.1002/oby.21681>
95. Kataoka J, Larsson I, Lindgren E, et al. Circulating anti-Müllerian hormone in a cohort-study of women with severe obesity with and without polycystic ovary syndrome and the effect of a one-year weight loss intervention. *Reprod Biol Endocrinol.* 2022;20(1):153. <https://doi.org/10.1186/s12958-022-01022-0>
96. Thomson RL, Buckley JD, Moran LJ, et al. The effect of weight loss on anti-Müllerian hormone levels in overweight and obese women with polycystic ovary syndrome and reproductive impairment. *Hum Reprod.* 2009;24(8):1976-1981. <https://doi.org/10.1093/humrep/dep101>
97. Kotlyar AM, Seifer DB. Ethnicity/race and age-specific variations of serum AMH in women-A review. *Front Endocrinol (Lausanne).* 2020;11:593216. <https://doi.org/10.3389/fendo.2020.593216>
98. Tal R, Seifer DB, Khanimov M, Malter HE, Grazi RV, Leader B. Characterization of women with elevated antimüllerian hormone levels (AMH): correlation of AMH with polycystic ovarian syndrome phenotypes and assisted reproductive technology outcomes. *Am J Obstet Gynecol.* 2014;211(1):59.e1-59.e8. <https://doi.org/10.1016/j.ajog.2014.02.026>
99. Jaffar M, Ahmad SN, Monica, Ashraf M, Shaik SA, Asif M. Geographical diversity in the age specific anti Müllerian hormone levels in infertile women: a hospital based cohort study. *J Hum Reprod Sci.* 2023;16(1):29-35. https://doi.org/10.4103/jhrs.jhrs_163_22
100. Kunicki M, Łukaszuk K, Liss J, Jakiel G, Skowrońska P. Demographic characteristics and AMH levels in rural and urban women participating in an IVF programme. *Ann Agric Environ Med.* 2018;25(1):120-123. <https://doi.org/10.26444/aaem/78944>
101. Emokpae MA, Brown SI. Effects of lifestyle factors on fertility: practical recommendations for modification. *Reprod Fertil.* 2021;2(1):R13-r26. <https://doi.org/10.1530/RAF-20-0046>
102. Tal R, Seifer DB. Potential mechanisms for racial and ethnic differences in antimüllerian hormone and ovarian reserve. *Int J Endocrinol.* 2013;2013:818912. <https://doi.org/10.1155/2013/818912>
103. Lie Fong S, Laven JSE, Duhamel A, Dewailly D. Polycystic ovarian morphology and the diagnosis of polycystic ovary syndrome: redefining threshold levels for follicle count and serum anti-Müllerian hormone using cluster analysis. *Hum Reprod.* 2017;32(8):1723-1731. <https://doi.org/10.1093/humrep/dex226>

104. Yue CY, Lu LK, Li M, Zhang QL, Ying CM. Threshold value of anti-Müllerian hormone for the diagnosis of polycystic ovary syndrome in Chinese women. *PLoS One*. 2018;13(8):e0203129. <https://doi.org/10.1371/journal.pone.0203129>
105. Dewailly D. Toward a universal serum antimüllerian hormone threshold as a surrogate for polycystic ovarian morphology on ultrasound: the story is not over.... *Fertil Steril*. 2021;116(4):1158-1159. <https://doi.org/10.1016/j.fertnstert.2021.08.005>
106. Mahboobifard F, Rahmati M, Ramezani Tehrani F. A single cut-off value of anti-Müllerian hormone should not be used for the diagnosis of PCOS in all reproductive-aged women. *Hum Reprod*. 2022;37(3):621-622. <https://doi.org/10.1093/humrep/deac012>
107. Kim JY, Tfayli H, Michaliszyn SF, Lee S, Nasr A, Arslanian S. Anti-müllerian hormone in obese adolescent girls with polycystic ovary syndrome. *J Adolesc Health*. 2017;60(3):333-339. <https://doi.org/10.1016/j.jadohealth.2016.10.015>
108. Pawelczak M, Kenigsberg L, Milla S, Liu YH, Shah B. Elevated serum anti-Müllerian hormone in adolescents with polycystic ovary syndrome: relationship to ultrasound features. *J Pediatr Endocrinol Metab*. 2012;25(9-10):983-989. <https://doi.org/10.1515/jpem-2012-0013>
109. Palomaki GE, Kalra B, Kumar T, *et al.* Adjusting antimüllerian hormone levels for age and body mass index improves detection of polycystic ovary syndrome. *Fertil Steril*. 2020;113(4):876-884.e2. <https://doi.org/10.1016/j.fertnstert.2019.12.012>
110. Barbagallo F, van der Ham K, Willemsen SP, Louwers YV, Laven JS. Age-related curves of AMH using the gen II, the picoAMH, and the elecsys assays in women with polycystic ovary syndrome. *J Clin Endocrinol Metab*. 2024;109(10):2561-2570. <https://doi.org/10.1210/clinem/dgae153>
111. Kushnir VA, Halevy N, Barad DH, Albertini DF, Gleicher N. Relative importance of AMH and androgens changes with aging among non-obese women with polycystic ovary syndrome. *J Ovarian Res*. 2015;8(1):45. <https://doi.org/10.1186/s13048-015-0175-x>
112. Moy V, Jindal S, Lieman H, Buyuk E. Obesity adversely affects serum anti-Müllerian hormone (AMH) levels in Caucasian women. *J Assist Reprod Genet*. 2015;32(9):1305-1311. <https://doi.org/10.1007/s10815-015-0538-7>
113. Pigny P, Gorisse E, Ghulam A, *et al.* Comparative assessment of five serum antimüllerian hormone assays for the diagnosis of polycystic ovary syndrome. *Fertil Steril*. 2016;105(4):1063-1069.e3. <https://doi.org/10.1016/j.fertnstert.2015.12.023>
114. Bungum L, Jacobsson AK, Rosén F, *et al.* Circadian variation in concentration of anti-Müllerian hormone in regularly menstruating females: relation to age, gonadotrophin and sex steroid levels. *Hum Reprod*. 2011;26(3):678-684. <https://doi.org/10.1093/humrep/deq380>
115. Fu YX, Wang H, Hu T, Wang FM, Hu R. Factors affecting the accuracy and reliability of the measurement of anti-Müllerian hormone concentration in the clinic. *J Int Med Res*. 2021;49(5):3000605211016161. <https://doi.org/10.1177/03000605211016161>
116. Pankhurst MW, Shorakae S, Rodgers RJ, Teede HJ, Moran LJ. Efficacy of predictive models for polycystic ovary syndrome using serum levels of two antimüllerian hormone isoforms (proAMH and AMH_{N,C}). *Fertil Steril*. 2017;108(5):851-857.e2. <https://doi.org/10.1016/j.fertnstert.2017.08.012>
117. Wissing ML, Mikkelsen AL, Kumar A, *et al.* Associations of different molecular forms of antimüllerian hormone and biomarkers of polycystic ovary syndrome and normal women. *Fertil Steril*. 2019;112(1):149-155.e1. <https://doi.org/10.1016/j.fertnstert.2019.03.002>
118. Bhide P, Gudi A, Shah A, Homburg R. Serum anti-Müllerian hormone levels across different ethnic groups: a cross-sectional study. *BJOG*. 2015;122(12):1625-1629. <https://doi.org/10.1111/1471-0528.13103>
119. Gupta M, Yadav R, Mahey R, *et al.* Correlation of body mass index (BMI), anti-Müllerian hormone (AMH), and insulin resistance among different polycystic ovary syndrome (PCOS) phenotypes - a cross-sectional study. *Gynecol Endocrinol*. 2019;35(11):970-973. <https://doi.org/10.1080/09513590.2019.1613640>
120. Oueslati I, Hammami MB, Boukriba S, *et al.* Anti Müllerian hormone as a diagnostic tool for polycystic ovary syndrome in women of reproductive age with morbid obesity. *Horm Mol Biol Clin Invest*. 2022;43(4):381-387. <https://doi.org/10.1515/hmbci-2021-0078>
121. Li X, Tang Q, Feng Y, Zhang Y, Tian W, Zhang H. The relationship between anti-Müllerian hormone and body composition components in patients with polycystic ovary syndrome of reproductive age. *Reprod Sci*. 2024;31(9):2843-2848. <https://doi.org/10.1007/s43032-024-01561-4>
122. Vagios S, James KE, Sacha CR, *et al.* A patient-specific model combining antimüllerian hormone and body mass index as a predictor of polycystic ovary syndrome and other oligo-anovulation disorders. *Fertil Steril*. 2021;115(1):229-237. <https://doi.org/10.1016/j.fertnstert.2020.07.023>
123. Md Muslim MZ, Mohammed Jelani A, Shafi N, *et al.* Correlation between anti-Müllerian hormone with insulin resistance in polycystic ovarian syndrome: a systematic review and meta-analysis. *J Ovarian Res*. 2024;17(1):106. <https://doi.org/10.1186/s13048-024-01436-x>
124. Li Y, Ma Y, Chen X, *et al.* Different diagnostic power of anti-Müllerian hormone in evaluating women with polycystic ovaries with and without hyperandrogenism. *J Assist Reprod Genet*. 2012;29(10):1147-1151. <https://doi.org/10.1007/s10815-012-9839-2>
125. Bhide P, Kulkarni A, Dilgil M, *et al.* Phenotypic variation in anti-Müllerian hormone (AMH) production per follicle in women with polycystic ovary syndrome (PCOS) and isolated polycystic ovarian morphology (PCOM): an observational cross-sectional study. *Gynecol Endocrinol*. 2017;33(10):801-806. <https://doi.org/10.1080/09513590.2017.1320377>
126. Romualdi D, Di Florio C, Tagliaferri V, *et al.* The role of anti-Müllerian hormone in the characterization of the different polycystic ovary syndrome phenotypes. *Reprod Sci*. 2016;23(5):655-661. <https://doi.org/10.1177/1933719115611751>
127. Bradbury RA, Lee P, Smith HC. Elevated anti-Müllerian hormone in lean women may not indicate polycystic ovarian syndrome. *Aust N Z J Obstet Gynaecol*. 2017;57(5):552-557. <https://doi.org/10.1111/ajo.12647>
128. Saxena U, Ramani M, Singh P. Role of AMH as diagnostic tool for polycystic ovarian syndrome. *J Obstet Gynaecol India*. 2018;68(2):117-122. <https://doi.org/10.1007/s13224-017-1066-4>

